# Hoja de Trabajo

1. Tabla transición

| δ | p | q | r | s |
|---|---|---|---|---|
| a | q | – | – | – |
| b | – | r | – | r |
| c | – | – | s | – |

$L = \{p, q, r, s\}$

$ab(cb)^*$

AFD

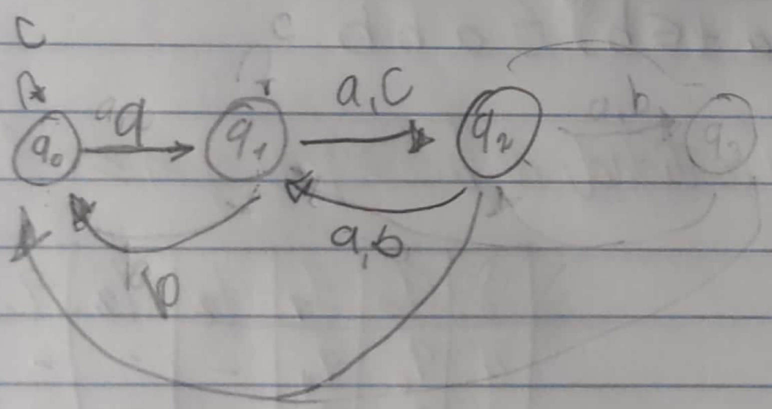
2.

$(((a+b)c^*(a+b)) + ((ac+ab)^*))^*$

$L = \{a, b, c\}$

| δ | $q_0$ | $q_1$ | $q_2$ |
|---|---|---|---|
| a | $q_1$ | $q_2$ | $q_1$ |
| b | – | $q_0$ | $q_1$ |
| c | $q_0$ | $q_2$ | – |

AFN



3. $L = \{\varepsilon, a, b\}$

| δ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| a | - | 3 | - | - | - | - | - | - |
| b | - | 5 | - | - | - | - | - | - |

$(\varepsilon)^* + (\varepsilon a + \varepsilon b)^* \varepsilon$

AFN

4. $L = \{\varepsilon, a, b\}$

| δ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| a | - | 3 | | - | - | - | - | 8 | - | - | - |
| b | - | 5 | - | - | 5 | - | - | - | 9 | 10 | - |

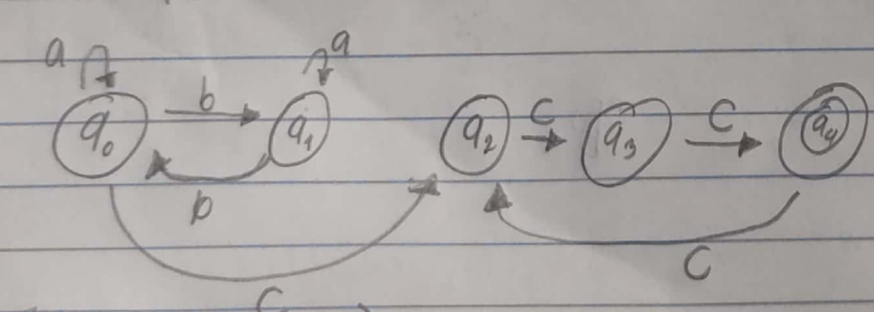
AFN

$(\epsilon)^* + (\epsilon a + \epsilon b)^* \epsilon abb$

5. $L_3 = \{ x c^{3m} / x \in \{a, b\}^*$ y la cantidad de b's es par y $m \geq 0 \}$

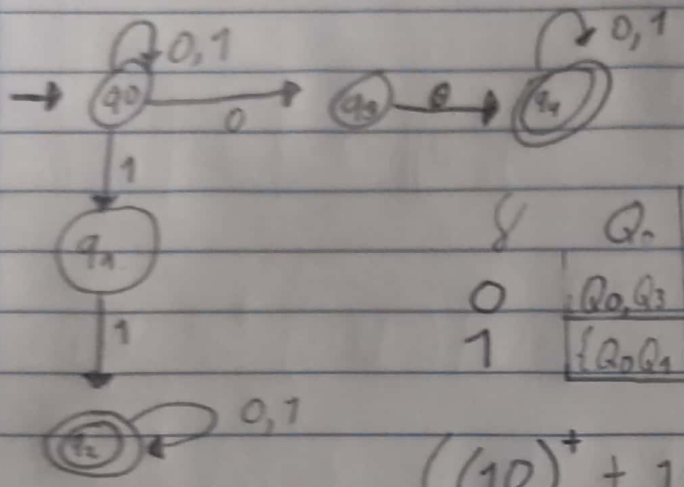
$L_3 = \{a, b, c\}$

| δ | $q_0$ | $q_1$ | $q_2$ | $q_3$ | $q_4$ |
|---|---|---|---|---|---|
| a | $q_0$ | $q_1$ | - | - | - |
| b | $q_1$ | $q_0$ | - | - | - |
| c | - | - | $q_3$ | $q_4$ | - |




$(a^+ b a^+ + b) c \quad + c^*$

6. $L_6 = \{ x / x \in \{0,1\}^*$ y x contiene la sub cadena 00 ó x contiene la subcadena 11 $\}$




| δ | $Q_0$ | $Q_1$ | $Q_2$ | $Q_3$ | $Q_4$ |
|---|---|---|---|---|---|
| 0 | $Q_0, Q_3$ | $Q_3$ | $Q_2$ | $Q_4$ | $Q_4$ |
| 1 | $\{Q_0 Q_1$ | $Q_2$ | $Q_2$ | $Q_4$ | $Q_4$ |

$$((10)^+ + 1 + (10)^+) + (0 + 01^+)$$